

ACTIVITY 03

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Visualization Activities with Datasets

Activity 1: Heatmap – Student Performance Analysis

Student	Mathematics	Physics	Chemistry	Programming	English
S1	92	88	84	95	81
S2	76	72	69	80	74
S3	81	78	83	85	79
S4	65	70	68	72	75
S5	90	94	91	96	89
S6	58	62	60	65	70
S7	84	86	82	88	80
S8	71	75	73	77	72
S9	95	91	93	98	94
S10	79	81	77	83	78

Questions:

1. Construct a Heatmap.
2. Identify highest overall performer.
3. Find highest average subject.
4. Identify students needing support.
5. Explain Heatmap advantages.

Python Code

Python

Run

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data = {
```

```
    "Student":["S1","S2","S3","S4","S5","S6","S7","S8","S9","S10"],
```

```

"Mathematics":[92,76,81,65,90,58,84,71,95,79],

"Physics":[88,72,78,70,94,62,86,75,91,81],

"Chemistry":[84,69,83,68,91,60,82,73,93,77],

"Programming":[95,80,85,72,96,65,88,77,98,83],

"English":[81,74,79,75,89,70,80,72,94,78]

}

```

```
df = pd.DataFrame(data)
```

```
df.set_index("Student", inplace=True)
```

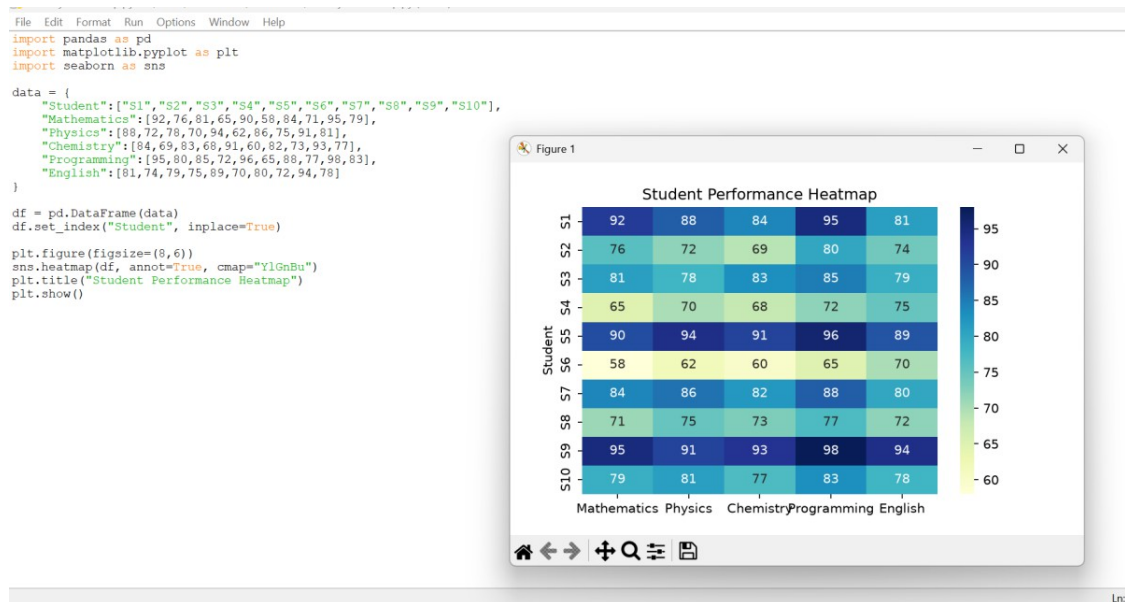
```
plt.figure(figsize=(8,6))
```

```
sns.heatmap(df, annot=True, cmap="YlGnBu")
```

```
plt.title("Student Performance Heatmap")
```

```
plt.show()
```

Output



Activity 2: Violin Plot & Boxplot – Employee Salary Analysis

Employee	HR	IT	Finance	Marketing
E1	4.2	6.5	5.8	5.1
E2	4.5	7.0	6.2	5.3

E3	4.7	7.4	6.0	5.5
E4	4.1	6.8	5.9	5.4
E5	4.6	8.2	6.7	5.8
E6	4.3	7.6	6.4	5.6
E7	4.8	8.5	6.9	5.9
E8	4.4	7.1	6.1	5.2
E9	4.9	8.8	7.2	6.0
E10	4.5	7.3	6.3	5.4

Questions:

1. Draw violin plots.
2. Construct boxplots.
3. Largest variation?
4. Highest median?
5. Which plot better represents distribution?

Python Code

Python

Run

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data = {
    "HR": [4.2, 4.5, 4.7, 4.1, 4.6, 4.3, 4.8, 4.4, 4.9, 4.5],
    "IT": [6.5, 7.0, 7.4, 6.8, 8.2, 7.6, 8.5, 7.1, 8.8, 7.3],
    "Finance": [5.8, 6.2, 6.0, 5.9, 6.7, 6.4, 6.9, 6.1, 7.2, 6.3],
    "Marketing": [5.1, 5.3, 5.5, 5.4, 5.8, 5.6, 5.9, 5.2, 6.0, 5.4]
}
```

```
df = pd.DataFrame(data)
```

```
plt.figure(figsize=(8,5))
```

```
sns.violinplot(data=df)
```

```
plt.title("Employee Salary - Violin Plot")
```

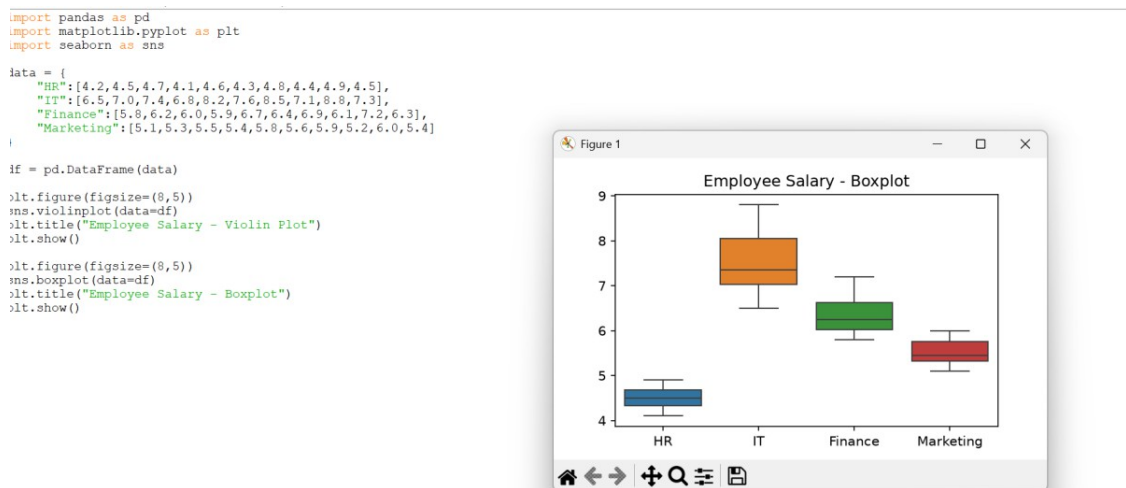
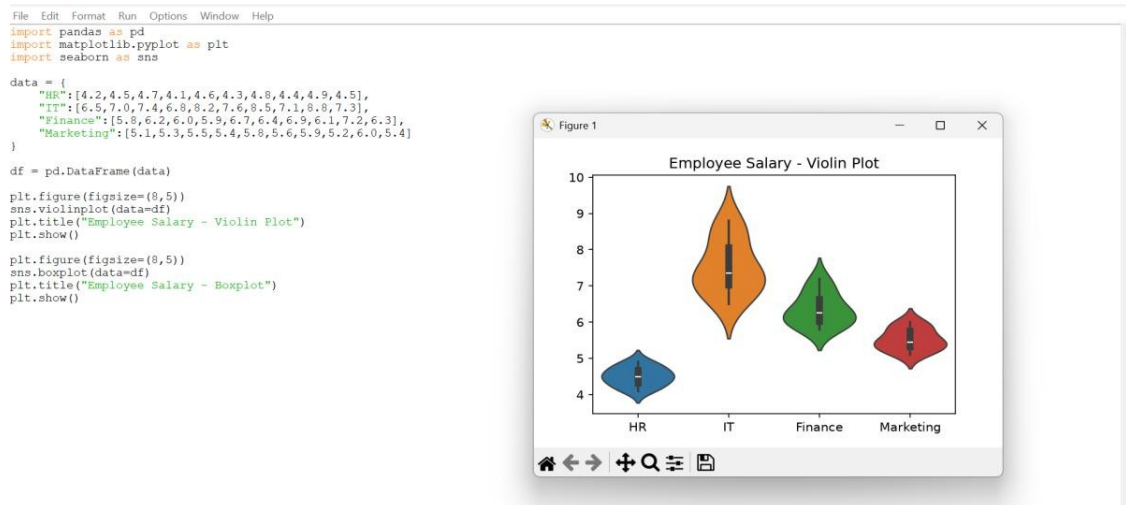
```
plt.show()
```

```
plt.figure(figsize=(8,5))
```

```
sns.boxplot(data=df)
```

```
plt.title("Employee Salary - Boxplot")
```

```
plt.show()
```



Activity 3: Histogram & Density Plot – Website Response Time

Obs	Response Time (ms)
-----	--------------------

1	210
2	225
3	230
4	240
5	250
6	260
7	270
8	280
9	285
10	290
11	300
12	305
13	310
14	315
15	320
16	330
17	340
18	350
19	360
20	380

Questions:

1. Draw histogram.
2. Plot density.
3. Determine skewness.
4. Common range.
5. Which plot better represents distribution?

Python Code

Python

Run

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
response_time = [210,225,230,240,250,260,270,280,285,290,
                 300,305,310,315,320,330,340,350,360,380]
```

```
df = pd.DataFrame({"Response Time":response_time})
```

```
plt.figure(figsize=(8,5))
```

```
plt.hist(df["Response Time"], bins=8)
```

```
plt.title("Histogram of Website Response Time")
```

```
plt.xlabel("Response Time (ms)")
```

```
plt.ylabel("Frequency")
```

```
plt.show() plt.figure(figsize=(8,5))
```

```
sns.kdeplot(df["Response Time"], fill=True)
```

```
plt.title("Density Plot of Website Response Time")
```

```
plt.show()
```

Output

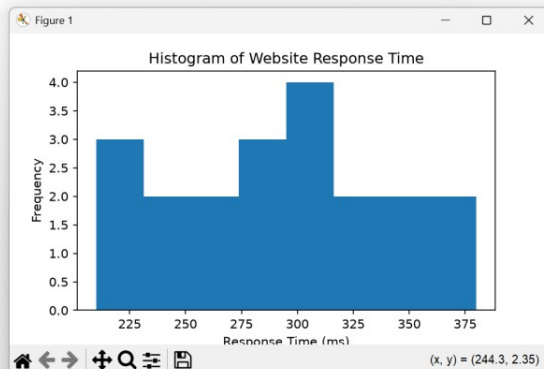
```
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

response_time = [210,225,230,240,250,260,270,280,285,290,
                 300,305,310,315,320,330,340,350,360,380]

df = pd.DataFrame({"Response Time":response_time})

plt.figure(figsize=(8,5))
plt.hist(df["Response Time"], bins=8)
plt.title("Histogram of Website Response Time")
plt.xlabel("Response Time (ms)")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(8,5))
sns.kdeplot(df["Response Time"], fill=True)
plt.title("Density Plot of Website Response Time")
plt.show()
```



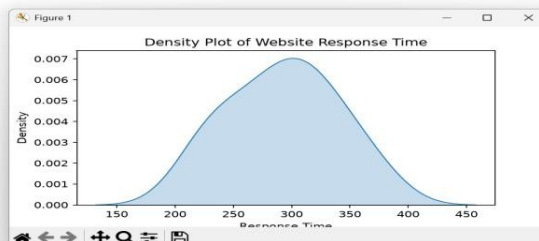
```
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

response_time = [210,225,230,240,250,260,270,280,285,290,
                 300,305,310,315,320,330,340,350,360,380]

df = pd.DataFrame({"Response Time":response_time})

plt.figure(figsize=(8,5))
plt.hist(df["Response Time"], bins=8)
plt.title("Histogram of Website Response Time")
plt.xlabel("Response Time (ms)")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(8,5))
sns.kdeplot(df["Response Time"], fill=True)
plt.title("Density Plot of Website Response Time")
plt.show()
```



Activity 4: Ridgeline Plot – Electricity Consumption

House	Jan	Feb	Mar	Apr	May
H1	220	235	250	275	295
H2	240	245	260	285	305
H3	230	238	255	280	300
H4	215	228	248	270	292
H5	225	240	258	282	298
H6	245	250	265	290	310
H7	235	242	259	284	302
H8	228	236	252	278	297
H9	238	246	262	287	307
H10	232	239	256	281	301

Questions:

1. Create ridgeline plot.
2. Highest average month?
3. Widest spread?
4. Describe trend.
5. Why ridgeline?

Python Code

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data = {  
    "Jan": [220, 240, 230, 215, 225, 245, 235, 228, 238, 232],  
    "Feb": [235, 245, 238, 228, 240, 250, 242, 236, 246, 239],  
    "Mar": [250, 260, 255, 248, 258, 265, 259, 252, 262, 256],  
    "Apr": [275, 285, 280, 270, 282, 290, 284, 278, 287, 281],  
    "May": [295, 305, 300, 292, 298, 310, 302, 297, 307, 301]  
}
```

```

df = pd.DataFrame(data)

plt.figure(figsize=(8,8))

offset = 0

for month in df.columns:

    sns.kdeplot(df[month], fill=True)

    plt.text(210, offset + 0.01, month, fontsize=10)

    offset += 0.02

plt.title("Ridgeline-style Plot of Electricity Consumption")

plt.xlabel("Electricity Consumption")

plt.ylabel("Density")

plt.show()

```

Output

File Edit Format Run Options Window Help

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Jan": [220, 240, 230, 215, 225, 245, 235, 228, 232],
    "Feb": [235, 245, 238, 228, 240, 250, 242, 236, 239],
    "Mar": [250, 260, 255, 248, 258, 265, 259, 252, 256],
    "Apr": [275, 285, 280, 270, 282, 290, 284, 278, 281],
    "May": [295, 305, 300, 292, 298, 310, 302, 297, 301]
}

df = pd.DataFrame(data)

plt.figure(figsize=(8, 6))

for month in df.columns:
    sns.kdeplot(df[month], fill=True, label=month)

plt.title("Electricity Consumption Distribution")
plt.xlabel("Consumption")
plt.ylabel("Density")
plt.legend()
plt.show()

```



Activity 5: Histogram, Density & Boxplot – Daily Steps

Person	20–30	31–45	46–60
P1	9500	8200	6500
P2	10200	8600	6700
P3	9800	8400	6900
P4	11000	9000	7200

P5	10500	8900	7000
P6	9700	8500	6800
P7	10100	8700	7100
P8	10800	9100	7300
P9	11200	9200	7400
P10	9900	8600	6950

Questions:

1. Draw histograms.
2. Plot density curves.
3. Create boxplots.
4. Highest median group?
5. Best visualization and justify.

Python Code

Python

Run

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

data = {

    "20-30":[9500,10200,9800,11000,10500,9700,10100,10800,11200,9900],

    "31-45":[8200,8600,8400,9000,8900,8500,8700,9100,9200,8600],

    "46-60":[6500,6700,6900,7200,7000,6800,7100,7300,7400,6950]

}

df= pd.DataFrame(data)

plt.figure(figsize=(8,5))

df.hist(figsize=(8,6))

plt.show()

plt.figure(figsize=(8,5))
```

for col in df.columns:

sns.kdeplot(df[col], label=col, fill=True)

plt.legend()

plt.title("Density Plot of Daily Steps")

plt.show()

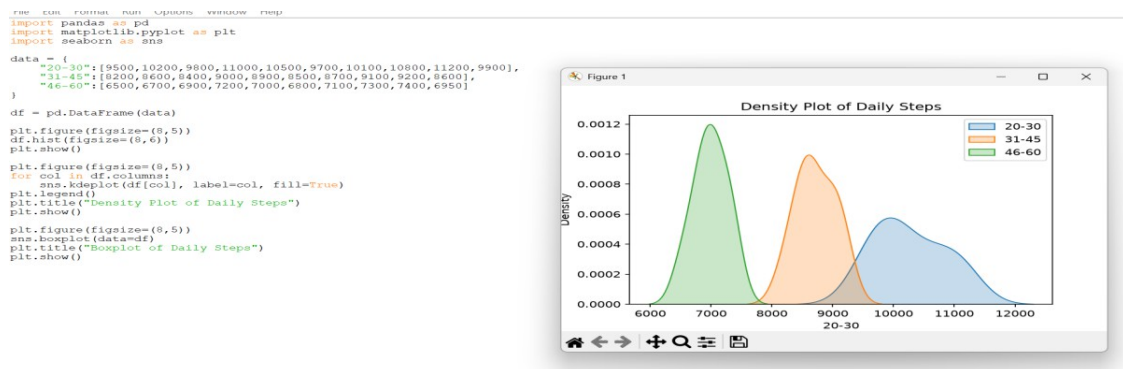
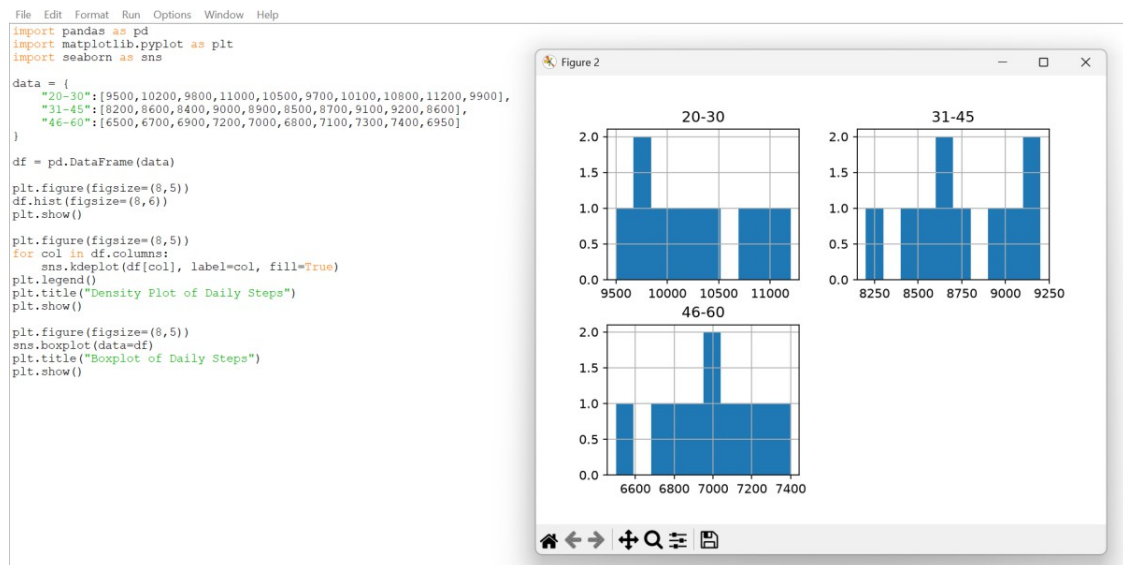
plt.figure(figsize=(8,5))

sns.boxplot(data=df)

plt.title("Boxplot of Daily Steps")

plt.show()

Output



File Edit Format Run Options Window Help

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "20-30": [9500, 10200, 9800, 11000, 10500, 9700, 10100, 10800, 11200, 9900],
    "31-45": [8200, 8600, 8400, 9000, 8500, 8700, 9100, 8200, 8600],
    "46-60": [6500, 6700, 6900, 7200, 7000, 6800, 7100, 7300, 7400, 6950]
}

df = pd.DataFrame(data)

plt.figure(figsize=(8,5))
df.hist(figsize=(8,6))
plt.show()

plt.figure(figsize=(8,5))
for col in df.columns:
    sns.kdeplot(df[col], label=col, fill=True)
plt.legend()
plt.title("Density Plot of Daily Steps")
plt.show()

plt.figure(figsize=(8,5))
sns.boxplot(data=df)
plt.title("Boxplot of Daily Steps")
plt.show()
```

